JOHN TAMBA KORFEH

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RESEARCH INTERESTS

Thermal Engineering • Computational Fluid Dynamics (CFD) • Renewable & Sustainable Energy Systems • Geothermal-Solar Hybrid Energy • Solar Thermal Power Generation • Heat Transfer & Energy Conversion • Thermodynamic Modeling

EDUCATION

B. Tech in Mechanical Engineering (Thermal Engineering) 2022 – 2026
Marwadi University, Rajkot, Gujarat, India

- Graduated First Class with Distinction; ranked Top 10% of Mechanical Engineering Department (120+ students)
- Completed 3 research projects yielding 2 conference presentations and 2 manuscripts under peer review
- Awarded Best Undergraduate Research Project (2026) for externally funded geothermal-assisted hybrid solar chimney capstone

Diploma in Automotive & Heavy Equipment Mechanics 2017 – 2021
Booker Washington Institute, Kakata, Liberia

- Graduated Magna Cum Laude; Most Decorated Graduate of Class of 2021 (16+ certificates and awards)
- Nominated as United World Colleges (UWC) Scholar, Liberia (2019)

RESEARCH EXPERIENCE & PUBLICATIONS

Geothermal-Assisted Hybrid Solar Chimney Power Plant Prototype [Ongoing] 2025 – Present
Undergraduate Researcher | Marwadi University | Externally funded by Dr. Alan Williams (UK)

- Designed, fabricated, and experimentally evaluated a geothermal-assisted hybrid solar chimney power plant prototype targeting affordable off-grid power generation for rural Sub-Saharan African communities
- Validated experimental data against CFD predictions (ANSYS Fluent) using thermocouple arrays, anemometers, and solar irradiance sensors for multi-parameter acquisition

Design & Development of a Hybrid Sustainable Power Generating Plant using Geothermal Energy-Numerical Analysis 2024 – 2025
Lead Researcher | Marwadi University | Externally funded by Dr. Alan Williams (UK)

- Developed coupled thermodynamic and CFD models using MATLAB, ANSYS Fluent, Workbench, and ICEM CFD to evaluate hybrid energy conversion efficiency across multiple operating scenarios
- Presented at FMFP Conference 2025 (100+ researchers); manuscript under peer review for journal publication

CFD Analysis of Aerodynamic Performance in a Quadcopter 2024 – 2025
Lead Researcher | Marwadi University

- Optimized lift-to-drag ratio and characterized turbulence behavior using k-ε and k-ω SST models in ANSYS Fluent across multiple angle-of-attack configurations
- Presented at TFGET Conference 2026; manuscript under peer review for journal publication

Research Intern — Fluid Power Engineering Lab May 2024 – Feb 2026
Dept. of Mechanical Engineering, Marwadi University | 1 year 10 months

- Conducted thermal-fluid system experiments and contributed to data analysis for faculty-led renewable energy research projects
- Gained hands-on experience with pressure transducers, anemometers, data collection instruments, irradiance meters, and thermal imaging for experimental fluid mechanics

Publication List:

[1] **Korfeh, J.T.** et al. "Design & Development of a Hybrid Sustainable Power Generating Plant Using Geothermal Energy." *FMFP 2025. Under review.*

[2] **Korfeh, J.T.** et al. "CFD Analysis of Aerodynamic Performance in a Quadcopter." *TFGET 2026. Under review.*

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Founder & CEO — Y-LEAD Liberia 2023 – Present
Youth for Leadership in Environment, Academics & Development

- Founded a youth-led NGO addressing environmental degradation, academic access, and socio-economic challenges affecting young Liberians
- Organized 5+ mentorship programs and environmental campaigns reaching 200+ students across multiple institutions
- Developed the organization's strategic vision connecting sustainable energy advocacy with youth empowerment

CEO — KDARC Study Abroad Agency 2024 – Present

- Built a mentorship organization that has guided 30+ Liberian students toward international university admissions and scholarship applications
- Provided one-on-one SOP review, university selection coaching, and application strategy for programs in India, Europe, and North America

International Student Representative — Marwadi University 2022 – 2026

- Advocated for 500+ international students and served as liaison between Freshers and university administration

INDUSTRY EXPERIENCE

CAT Service Technician — Mantrac Liberia, Monsterrado County Oct 2021 – Nov 2022

- Diagnosed and repaired CAT heavy equipment (excavators, loaders, dozers) using Caterpillar Electronic Technician (ET) and CAT Service Information System (SIS); performed hydraulic, mechanical, and electrical troubleshooting; certified to operate CAT Forklift, Wheel Loader, and Dozer

Automotive Mechanic — VA & Sons Automotive, Monrovia 2014 – 2022

- Accumulated 7+ years of hands-on experience in vehicle diagnostics, engine repair, and heavy machinery maintenance, building practical understanding of thermodynamic and fluid power systems

TECHNICAL SKILLS

Simulation & CFD: ANSYS Fluent, ANSYS Workbench, ICEM CFD, MATLAB/Simulink
Design & CAD: SolidWorks, AutoCAD
Research Methods: CFD Turbulence Modeling (k-ε, k-ω SST), Thermodynamic Modeling, Experimental Validation, Prototype Fabrication, Data Acquisition
Practical: Heavy Machinery Diagnostics, Hydraulic Systems, Engine Repair, CAT Equipment Operation
Languages: English (Native/Fluent)

HONORS & AWARDS

- Best Undergraduate Research Project — Mechanical Engineering Dept., Marwadi University (2026)
- Best Research Poster, 2nd Runner-Up — ASME Annual Conference, India Chapter (2025)
- Young Innovator Award, Top 100 Asia-Pacific — Global Peace Foundation Japan (2025)
- Most Decorated Graduate — Booker Washington Institute, Class of 2021 (16+ certificates)
- Community Social Impact Award — Omega Old Field Block C, Liberia (2020)
- UWC Scholar Nominee — Liberia National Selection (2019)

CERTIFICATIONS

- ANSYS Simulation Training — Marwadi University (2025)
- NPTEL Engineering Metrology — IIT Kanpur (2024)
- MATLAB for Mechanical Engineers — Internshala Trainings (2024)
- General Caterpillar Technician/Mechanic — Technician for Africa, Magna Cum Laude (2021)

PROFESSIONAL MEMBERSHIPS

American Society of Mechanical Engineers (ASME) • Institute of Electrical and Electronics Engineers (IEEE) • Engineering Society of Liberia

COMMUNITY ENGAGEMENT & SERVICE

- Secretary, Community Watch Team — Coordinated grassroots safety and awareness programs serving 500+ residents in underserved neighbourhoods, Monrovia (2019–2022)
- Environmental Activist — Championing sustainable energy access and environmental literacy through Y-LEAD campaigns and community workshops across Liberia
- Peer Mentor — Supported incoming international students at Marwadi University with academic adjustment and cultural integration.